FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

First Semester

Multi-Disciplinary Course - MG1MDCCFS101 - DATA RECOVERY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 35

SCHEME (ANSWER KEYS)

Part A

Short Answer Type Questions Answer any 10 questions

Each question carries 1 marks (10 × 1 = 10)

1. The primary function of a hard disk drive's (HDD) read/write head is to read data from and write data to the disk's platters.

2. Revolutions Per Minute

3. Data loss refers to the unintended and irreversible loss or destruction of digital information stored on a device or storage medium. Mention Human error, Hardware failure, Software issues, Malware and viruses

4. Firmware is a specialized type of software that is embedded into hardware devices to control and manage their functions. Firmware is typically stored in non-volatile memory, such as ROM, EEPROM or flash memory, ensuring that it persists even when the device is powered off.

5. A hard disk drive (HDD) is a data storage device that uses magnetic storage to store and retrieve digital information.

Explain briefly about the primary elements of a hard drive's structure: Platters, Spindle, Read/Write heads, Actuator Arm, Tracks and sectors.

6. Rotational latency is the time delay for the desired sector of a spinning disk to align under the read/write head in an HDD.

7. Replication and Propagation: Malicious viruses can replicate themselves and spread across systems, often exploiting vulnerabilities or user actions like opening infected files or links.

Payload Delivery: Viruses can deliver harmful payloads, such as deleting data, stealing information, or compromising system functionality.

8. Explain briefly either two of the following tools Recuva / Easeus/ Photorec/ Disk drill/ R-studio

9. R-Studio is a comprehensive and professional data recovery tool, suitable for both home users and data recovery specialists. It is highly effective for recovering data from a variety of file systems and offers advanced recovery features.

10. Raw Recovery: FDAC excels at raw data recovery and file carving, scanning disks at a lower level (sector by sector) and recovering files by their signature patterns, even when the file system is damaged or deleted.

• File Signature Search: It allows users to search for specific file types (e.g., JPEG, PDF, DOCX, MP4) based on their signatures.

• Works with Multiple File Systems: FDAC supports various file systems, including FAT, NTFS, ext2/ext3/ext4, and more.

11. The typical file extension for disk images is .iso

12. Common causes of accidental deletion of data include human error and software malfunctions.

Part B

Short Essay Type Questions Answer any 3 questions

Each question carries 5 marks (3 × 5 = 15)

13. The actuator arm is a key component in the hard disk drive architecture,. It is responsible for positioning the read/write heads over the correct track on the disk platters to access or store data and enabling efficient data access and operation of the HDD.

Explain Key Functions of the Actuator Arm:

Positioning the Read/Write Heads, Movement Across the Platters, Controlled by the Voice Coil Motor (VCM), Fast and Accurate Positioning, Handling Multiple Platters, Minimizing Latency

Explain the Key function - Mark 4

Explain HDD Figure - Mark 1

14**. Explain 5 difference : Mark 5**

Parameters	ANTIVIRUS	ANTIMALWARE
Definition	Antivirus is a software program that protects the computer system from viruses.	Antimalware is a software program that protects the computer systems from all malware i.e. viruses, trojans, worms, etc.

Protects	It protects from traditional viruses.	It protects from all malware including newer and sophisticated ones.
Danger	It protects from predictable danger.	It protects from unpredictable danger.
Used in	It is mostly used in personal computers.	It is mostly used in organizational computers.
Cost	It is comparatively less costly.	It is more costly.
Detect	It detects and destroys viruses which is a type of malware.	It detects and destroys all malware including viruses too.
Features	• Real time scanning • Remove threats	• Sandboxing • Traffic filtering • Proactive Security
Techniques	Techniques Used by Antivirus software- • Scanning • Integrity checking • Interception	Used by Anti-malware software- • Sandboxing • Signature-based detection • Behavior-based detection

15. exFAT is a file system introduced by Microsoft as an improved version of FAT32. It addresses some of the limitations of FAT32, allowing for larger file sizes and better performance. exFAT is commonly used for removable storage devices, such as external SSDs, hard drives and SD cards

Explain 5 Features:

1 Large file and partition size support

2 Efficient disk space utilisation

3 Compatibility: It can natively be read from and written to by Windows and MacOS operating systems without the need for third-party software.

4 Efficient for Flash Storage: exFAT is optimized for use on flash memory devices like USB drives and SD cards. It uses less system overhead than NTFS.

5 exFAT improves performance over FAT32 with features like better allocation of free space and faster read/write operations.

Definition : Mark 1

Features : Mark 4

16. RDrive is a powerful tool that you can use to create a disk image of your hard drive - a duplicate or backup copy of everything on the drive. With it, it's possible to create logical disks and partitions without interrupting your operating system at work. The resulting disk images can then be stored on extractable units, like CDs or DVDs.

• R-Drive can restore the image in the original disk, in any of its partitions, or even in free space.

Explain the Uses of R Drive: Disk Cloning, Data Backup, Disk Restoration, Disaster Recovery, File and Partition Management

Definition ; Mark 2

Explain the uses of R Drive : Mark 3

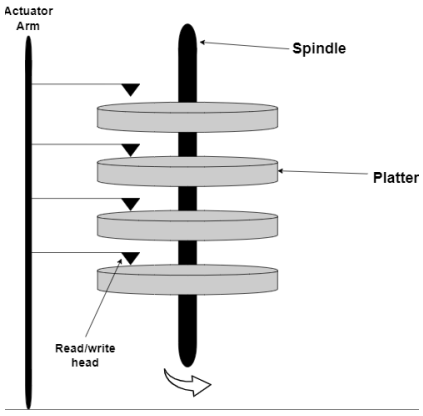
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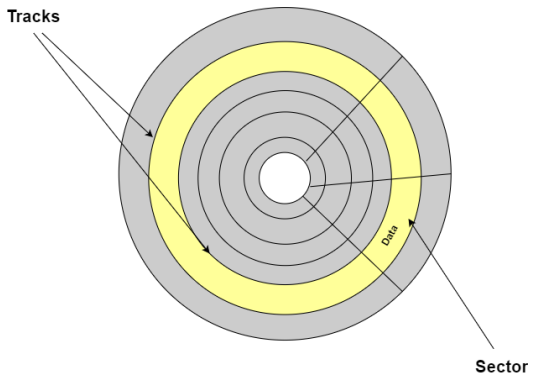
Essay Type Questions Answer any 1 questions

Each question carries 10 marks (1 × 10 = 10)

17. A **hard disk drive (HDD)** consists of several key components that work together to store and retrieve data efficiently. Here is an explanation of the various components of a hard disk. Explain each physical components of hard disk:

• Platters , Spindle, Read/write heads ,Tracks, Sectors





Definition of Hard disk – Mark 2

Explain architecture – Mark 3

Explain the figure -Mark 5

18. Disk partitioning is the process of dividing a physical hard drive (or storage device) into multiple logical sections, known as partitions. Each partition can be treated as a

separate disk, and they can have different file systems. Partitioning is commonly done to organize data, improve performance, or install multiple operating systems on a single disk. A disk consists of the following parts:

MBR (Master Boot Record) or GPT (GUID Partition Table)

Primary Partition: The main partition where an operating system or data is stored.

Extended Partition: A special partition that can be divided into logical partitions

Explain the steps how to create new disk Partition:

1. Back Up Important Data:

2.Access the Partitioning Tool:

In Windows: Right-click on "This PC" > "Manage" > "Disk Management."

In Linux: You can use GParted or command-line tools like fdisk or parted.

3.Select the Disk to Partition:

4.Delete Existing Partitions (if necessary):In Disk Management (Windows): Right-click on the existing partition > "Delete Volume."

5. Create New Partitions:Example: You can create a primary partition for the OS (e.g., C: drive) and an extended partition for other data storage. The extended partition can be further split into logical partitions (e.g., D: and E:).

6.Specify Partition Size and File System:(e.g., NTFS, FAT32, ext4, exFAT).

7.Apply Changes

8.Format the Partitions: Format each partition with the chosen file system.

In Windows: Right-click on the partition > "Format" > Choose the file system.

9. Assign Drive Letters: Eg: C:, D:, E:.etc, can be assigned to the primary and logical partitions

Explain Definition of Disk Partition : Mark 3

Explain the steps to create a disk partition : Mark 7